

AI Persona — Evaluation Report

Scaler AI Engineer Screening Assignment · Candidate: Chandu Bhargav Adithya · June 2026

1. Voice Quality

Latency was logged server-side on every call using `time.time()` at webhook entry and reply generation. Transcription accuracy was evaluated over 10 test phrases comparing Deepgram Nova-2 output against manually typed ground truth. Booking success was recorded across 10 end-to-end test calls.

Metric	Result	Measurement Method
First response latency — average	1.8 s	Railway logs: <code>t_start</code> to reply generation
First response latency — minimum	1.2 s	Short factual queries
First response latency — maximum	3.1 s	Complex project description queries
STT transcription accuracy	94.2 %	10 test phrases, manual comparison
Booking task completion rate	100 %	Calendly link or slot confirmed in all 10 calls
Barge-in / interruption handling	Stable	Handled natively by Vapi — no crashes observed

2. Chat Groundedness & Retrieval Quality

Hallucination rate was measured using a judge model (Gemini 2.5 Flash) against a 12-question golden Q&A; set spanning resume, GitHub projects, out-of-scope queries, and adversarial prompt injections. Each answer was scored 0.0 (fully grounded) to 1.0 (hallucinated). Retrieval precision was measured as the fraction of expected keywords present in the top-5 retrieved chunks per query.

Category	Questions	Hallucinated	Rate	Retrieval Precision
Resume / Background	4	0	0.0 %	0.89
GitHub Projects	5	0	0.0 %	0.91
Out-of-scope	1	0	0.0 %	N/A
Adversarial / Injection	2	0	0.0 %	N/A
Overall	12	0	0.0 %	0.90

3. Failure Modes

Failure Mode	Root Cause	Fix Applied
Bot returned 'not in knowledge base' for all project queries on Railway	Vector DB was in <code>.gitignore</code> and never rebuilt on Railway. Ingest script did not run during deployment.	Added <code>python -m rag.ingest</code> to Railway build phase via <code>nixpacks.toml</code> .
Project details mixed between repos — wrong info returned	Pure semantic search returned chunks from similar repos. LandCoverClassification and lulc-dl both cover land cover topics.	Implemented repo-aware retrieval: query keywords trigger direct metadata filtering.

Voice responses too long with spoken markdown symbols	Prompt did not distinguish voice from chat. Bullet points and bold markers were spoken literally by TTS.	Added is_voice=True flag. Voice mode caps at 300 tokens and 3 sentences.
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4. Conscious Tradeoff — Coverage vs. Latency

The repo-aware retrieval strategy fetches all chunks from a matched repository by metadata filter rather than relying solely on semantic similarity scores. This improved retrieval precision from approximately 0.60 to 0.90 but adds roughly 200 ms per query due to scanning all 2,638 stored chunks for metadata matches.

Coverage was deliberately prioritised. In a hiring context, a factually wrong or missing answer about a candidate's project is more damaging than an additional 200 ms of latency. The resulting average end-to-end latency of 1.8 s still satisfies the assignment requirement of under 2 s for first response.

5. What I Would Build With 2 More Weeks

Improvement	Description
Streaming responses	Stream Gemini tokens directly to the chat UI for perceived sub-second latency rather than waiting for full generation.
Google Calendar integration	Replace Calendly with Google Calendar FreeBusy API — agent proposes specific slots, creates events, and sends confirmations automatically.
Retrieval confidence scoring	Attach a confidence score to each answer. If the top chunk distance exceeds a threshold, the agent explicitly flags uncertainty.
Automated eval pipeline	Run the evaluation suite on every git push via GitHub Actions to catch retrieval regressions before they reach production.
Persistent conversation memory	Store chat history in Redis so returning users resume where they left off across sessions.

All systems live and tested at submission. Voice agent callable 24/7 via Vapi phone number. Chat interface publicly accessible at Railway URL.